FYP-2 Mid Evaluation — 5-Minute Speaking Script

S26-017-R-GASR · FAST NUCES Islamabad · Supervisor: Dr Akhtar Jamil

Written against the finalised 24-slide deck. Slides 1–7 are the talk, 8–12 the demonstration, 13–24 held for questions.

Member	Slides	Words	Time	Section
Ahmad Rashad 22i-1175	1 – 3	228	1:38 – 1:49	Goals and objectives · faculty feedback addressed
Shehroz 22i-0771	4 – 6	309	2:12 – 2:28	Progress: methodology, what we measured, the base papers
Humair Khan 22i-2623	7	86	0:37 – 0:41	Remaining work, then drives the live demonstration
Total		**623**	**4:27 – 4:59**	Counted from the spoken lines only. Fast end 140 wpm, slow end 125 wpm.

Two rules this script follows. (1) Numbers stay on the slides. A figure is spoken aloud only when it means something on its own — "all ten videos", "one camera instead of two". Error scores, correlations and decibels are on screen where the panel can read them at their own pace, and are described in words instead. **(2) Nothing is read off a slide.** The deck carries the detail; you carry the argument.

Hard constraints from the committee. The 4–5 minutes are **uninterrupted** — do not stop for questions during slides 1–7, and do not overrun; every minute borrowed is a minute taken from the demonstration. Everything on **one laptop**. The demonstration must be the **live system**, not a video.

Ahmad Rashad (22i-1175) — Slides 1 to 3

Slide 1 — From Reconstruction to Anticipation · 28 words · 12 s

Good morning. I'm Ahmad Rashad, with Shehroz and Humair. Our supervisor is Dr Akhtar Jamil. Project S26-017 — a research FYP on 4D surgical reconstruction and deformation prediction.

Slide 2 — What the project is for · 104 words · 45 s

A surgeon operating through one laparoscope sees a flat image of the present moment. Our goal is a system that reconstructs that scene in 3D and anticipates how the tissue will deform next — from one camera, with no depth sensor and no annotation.

Point along the four cards. Say every status — the panel is marking progress.

Four objectives. Reconstruct in 3D from a single camera: complete, that was FYP-1. Learn how tissue deforms, without labels: complete. Predict forward with its own confidence: implemented. Evaluate against the literature: in progress.

Pause. Read the question slowly — it is the sentence they should remember.

So the research question is this. Can future tissue geometry be predicted from ordinary surgical video — and can the system tell us when to trust it?

Slide 3 — Faculty feedback from FYP-1 · 96 words · 41 s

Say the feedback, say what was built. The file names are on the slide — do not read them out.

We had four pieces of feedback after FYP-1, and all four are implemented.

Implement a base paper for comparative analysis — we implemented two; Shehroz will come back to them.

Implement state-of-the-art methods — we built eleven, classical and learned, all measured the same way on the same data.

Do a detailed tuning and optimisation analysis — that is Module 8.

And begin a paper in conference format — there is an IEEE manuscript in the repository.

All four are written. Two still need their compute runs — the first item on our remaining-work slide.

Step back from the laptop.

Shehroz.

Shehroz (22i-0771) — Slides 4 to 6

Slide 4 — The methodology, finalised · 90 words · 39 s

Thank you, Ahmad. The methodology is fixed. We track points on the tissue surface, look at the last few frames of how they moved, and predict where they go next. Nothing in that chain needs a human to label anything.

The model is a diffusion model — the family used for image generation, but generating motion rather than pictures.

The middle card. This is the one decision worth naming.

One decision matters most. Rather than predicting the motion outright, it predicts the *correction* to a simple straight-line guess — so even when it learns little, it still beats guessing naively.

Slide 5 — Metrics chosen, and what they measured · 102 words · 44 s

Lead with the three questions, not the metric names. The figures are on the slide.

We measure three things: is the shape in the right place, is the motion pointing the right way, and does the model know when it is unsure.

The second matters because a shape score alone cannot separate a model that learned nothing from one that learned the right direction but the wrong distance.

Three answer cards, right side. Say what each means, not its value.

The answers. On accuracy we are not yet ahead of the simple baselines, and we lose on all ten videos. But the direction is clearly learned, and the model's confidence tracks its real error. So the failure is *how far*, not *which way* — a far more tractable problem.

Slide 6 — What we build on, and where we go past it · 117 words · 50 s

The comparison slide, and it deserves its time. Four rows, left then right.

Four bodies of work sit next to ours.

EndoGaussian and Deform3DGS reconstruct surgical scenes beautifully — but with two cameras, and only what already happened. We reach comparable quality from one camera, then predict forward.

PointRNN is the closest published method to ours, and our main learned baseline — but built for rigid outdoor scenes, and it gives one answer with no confidence.

Luo and Hu built the first diffusion model over 3D points — but it generates shapes, not futures. Ours makes that idea conditional on time.

And surgical anticipation does predict the future — but in flat pixels, and only when told the instrument's action.

Read the gold line as your closing sentence.

No published method does all of it at once.

Step back.

Humair.

Humair Khan (22i-2623) — Slide 7, then the demonstration

Slide 7 — Remaining work until the final evaluation · 86 words · 37 s

Give each item a status, not a hope.

Five things remain, in the order they have to happen.

One: run the full comparison and the tuning studies. Both are written and tested — what is left is compute, not code.

Two: evaluate the two corrections already implemented. One of them needs no retraining at all.

Three: hold out an entire video, rather than frames from videos we also train on.

Four: fold the finished results into the manuscript.

Five: submit, and grow the dataset.

Go straight into the demo. Do not pause for questions.

And with that, let me show you the system running.

The Demonstration — Humair drives, slides 8 to 12

No verbatim script, because a live system will not run to one. These are the lines to have ready while cells execute.

Slide 8 — while the notebook loads

Everything from here is live — nothing is pre-rendered. We load the trained model, check the data it is being tested on, predict a moment it has never seen, draw that prediction as an actual video frame, compare it against the real one, and then run past the end of the recording.

Slide 9 — while the render cell runs

There is a gap to bridge here. The prediction moves a few hundred tracked points; a picture needs the whole scene, which is tens of thousands. So the motion of the tracked points is carried outward onto everything around them.

Two honest notes about what you are seeing. Tissue moves only a pixel or two between frames, which is invisible, so the arrows are magnified — and that magnification is printed on the panel and never applied to any measurement. And the forecast runs past the end of the recording: the real-video panel stops and says so, while the predicted panel keeps going. That is the only output that shows prediction rather than describing it.

Slides 10 to 12 — describing the results

Agree beforehand whether you or Shehroz narrates these. One of you, not both.

The key finding: the model knows which way the tissue moves, not how far. Then uncertainty — when the model's own samples disagree with each other, its prediction really is worse, which means it can flag its own weak moments. A system that is confidently wrong is unusable in theatre; one that says "don't trust me here" is usable even when imperfect. And finally the full comparison against both simple baselines.

> **Demo checklist — the day before, not on the day.**
>
> - Runtime connected, GPU confirmed, checkpoint verified loadable.
> - Run the whole demo path end to end twice. Time the slowest cell; know what you will say over it.
> - Deck, notebook, checkpoint and outputs on **one laptop**.
> - Fallback: if the runtime dies, show the already-executed notebook and say plainly that it is not re-executing. Do not pretend.
> - Screen resolution set so notebook text is readable from the back of the room.

Questions — and where the numbers live

Because the talk is deliberately light on figures, expect a question asking for them. That is fine: every figure is on a slide. Do not recite from memory — turn to the slide.

If the question is about…	Answers · slide
The network itself, why diffusion, the forward pass, conditioning, training, overfitting	**Ahmad** · 14 (architecture), 15, 16, 19

Goals, scope, how the FYP-1 feedback was addressed	**Ahmad** · 2, 3
Metrics, baselines, the quality tests, evaluation integrity, the comparison numbers, the direction finding, uncertainty	**Shehroz** · 5, 10–12, 17, 18
Literature positioning, the base papers, reconstruction against published figures	**Humair** · 20, 23
The demo, rendering, the forecast, the dataset, the pipeline	**Humair** · 9, 21, 22

Slide 14 is now a full architecture figure. It shows one forward pass of the denoiser as it is actually implemented — the two encoders, the temporal transformer over the four-frame axis, the horizon embedding injected into both branches, the two cross-attention blocks, and the output head — then the checkpoint crossing into Module 6 and the sampler, the three query modes and the baselines. If you are asked how the model works, go there and trace it left to right. Ahmad should be able to walk every box on it.

The one answer to agree on. "Your model doesn't beat a simple baseline — why should this pass?" One person answers: *"Because we can tell you exactly why it doesn't. We measured that it learns which way tissue moves and fails only on how far; we tested and ruled out the two obvious alternative explanations; and we have implemented the two corrections that follow. We also get a confidence estimate that none of the baselines can give you. For a research project, a measured negative with an identified cause is worth more than an unexplained positive."*